LEWIS MOTEN

Selected Projects | Front-End Engineering, Applied AI & Creative Tools

Public work and live demos with source links where available.

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Applied AI and visualization



Local Agent Lab | Portable AI workspace

Built a React and TypeScript browser workspace for coding, research, and scheduled agents. Runs small models locally or connects to Ollama and API providers, with project files and conversations stored in a browser folder or on a removable drive.

Live demo | Source



Zogograph Client | AI-assisted server monitoring

PHP/SQLite collectors and a JavaScript dashboard make server telemetry, alerts, and diagnostics readable. An AI assistant calls diagnostic tools and tracks the cost of an investigation.

Source | Videos



Crystal-9 | Tiny MoE for 8-bit hardware

Redesigned a sparse MoE tic-tac-toe policy for a 13-token vocabulary and tight memory budgets. Released a 46,299-byte packed INT4 artifact for a custom local runtime, with zero policy misses across 294,778 legal histories. A compact adaptation powers Crystal Palace 9 on the C64.

Source | Hugging Face



D64 Storage Lab | Repair and optimization

A JavaScript workbench maps Commodore 64 disk sectors and file chains, estimates read efficiency, repairs corruption, restores deleted files, and compares optimized and fragmented layouts.

Live demo | Source | Video

Browser tools and public systems



Sculpted Prim Toolkit | Three.js editor

Decodes Second Life-style RGB sculpt maps into editable 3D geometry. Camera views, mesh overlays, textures, and vertex tools support PNG, glTF/GLB, OBJ, and STL exports.

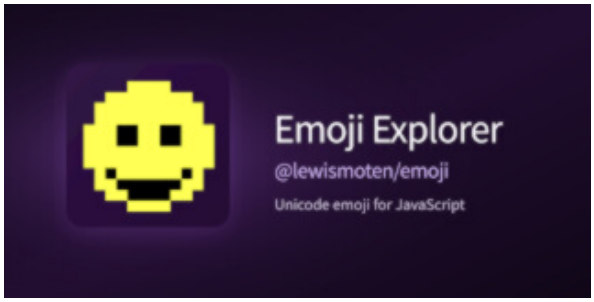
Live demo | Source | Videos



Data Over Audio | Web Audio API

Transfers text and binary data through sound using MFSK modulation, packetization, CRC checks, Hamming error correction, interleaving, and repeat requests.

Live demo | Source | Videos



Emoji Explorer | Unicode and pixel font

A localized Unicode explorer with version and sequence filters, favorites, keyboard navigation, and offline PWA support. Includes a custom 12 x 12 Pixel Emoji font and pixel editor.

Live demo | Source | npm



Dreamy Tags | WordPress plugin

An open-source PHP/Gutenberg plugin for category- and tag-filtered clouds, with exclusion rules and minimum-use thresholds. Published as a block, widget, and shortcode.

Plugin directory | Source

Interactive and creative projects



Restoring the Signal | Dialable exhibit

A live installation at Stone Branch Center for the Arts joins Raspberry Pi, Asterisk, and a PHP contribution site. It handles telephone audio, dialplans, transcription, and TTY support.

Live demo | Source | Videos | Press and photos



QR Code Generator | Creator and inspector

A browser tool backed by a first-party ES module for mixed segmentation, versions 1-40, Reed-Solomon correction, mask selection, and structural inspection.

Live demo | Source



Tiny Pockets Press | Miniature publishing

A modular JavaScript editor for miniature books, with layout and imposition logic and separate interior and cover print/PDF views.

Live demo | Source | Videos



Crystal Palace 9 | AI on 8-bit hardware

Built a Commodore 64 tic-tac-toe game around Crystal-9, a compact mixture-of-experts model trained to choose moves. Adapted inference and a command-room interface to the machine's tight memory and 8 x 8 multicolor graphics constraints.

Source | Video